

SLIIT Faculty of Computing

Year 2 Semester 1 - 2026

Final Group Report

IT2011 - Artificial Intelligence and Machine Learning

Project: Emotion-Aware Movie Recommendation System

Task: Seven-class emotion classification and recommendation

Academic year: Year 2 Semester 1 - 2026

Emotion-Aware Movie Recommendation System

Final Report

Executive Summary

This project developed an end-to-end NLP classification and recommendation workflow using the assigned movie dataset. Six independently tuned model families were evaluated on a common seven-class description-level dataset. Linear SVM achieved the highest held-out macro F1 (0.2523) and accuracy (0.3641) and was selected for the recommendation demonstration. Performance remains limited by class imbalance, subjective labels, and a small number of independent descriptions.

Student ID	Student	Assigned model
IT25100112	Karunaratne H. L. N. A.	Logistic Regression
IT25100083	Unawatuna H. M. I. D.	Linear Support Vector Machine
IT25100078	Thujikoshan Y.	Multinomial Naive Bayes
IT25100090	Hettiarachchi H. C. N. G.	Decision Tree
IT25100111	Chandrasekara C. M. T. D.	Random Forest
IT25100149	Rajapaksha P. A. D. S.	Multi-Layer Perceptron

1. Introduction and Problem Statement

Users may wish to find movies that match their current mood, while conventional recommenders focus on popularity, ratings, and behavior. The proposed solution predicts emotion from a movie description and combines predicted emotion with genre, rating, and popularity signals to rank catalogue recommendations.

2. Dataset Description

The raw CSV contained 46,173 rows and eight columns. Auditing identified repeated reviews with inconsistent targets, so one unique, label-consistent movie description became the modelling unit. The processed full EDA dataset contains 2,060 descriptions and eight emotions. The modelling dataset contains 2,057 descriptions and seven classes after excluding three surprise observations.

3. Preprocessing and EDA

Stage	Group implementation
Quality audit	Validated schema, nulls, blanks, rating range, exact duplicates, description uniqueness, and conflicting labels.
Missing genres	Encoded 35 empty genre lists using the explicit unknown_genre category.
Aggregation	Created one independent record per description and aggregated rating and review statistics.
Feature engineering	Text lengths, genre count, rating spread, review counts, unique counts, and log popularity.
Encoding	TF-IDF for descriptions and one-hot encoding for genres.
Outliers and scaling	Fold-fitted IQR clipping plus appropriate numerical scaling.
Selection and reduction	Chi-squared feature selection and TruncatedSVD where dense inputs were needed.
Leakage control	All learned transformations were fitted inside pipelines on training folds only.

EDA showed severe target imbalance, overlapping rating/length distributions, skewed popularity, genre imbalance, and vocabulary overlap. These findings motivated stratification, macro-F1 optimization, log transformation, class-aware tuning, and text-focused modelling.

4. Model Design and Implementation

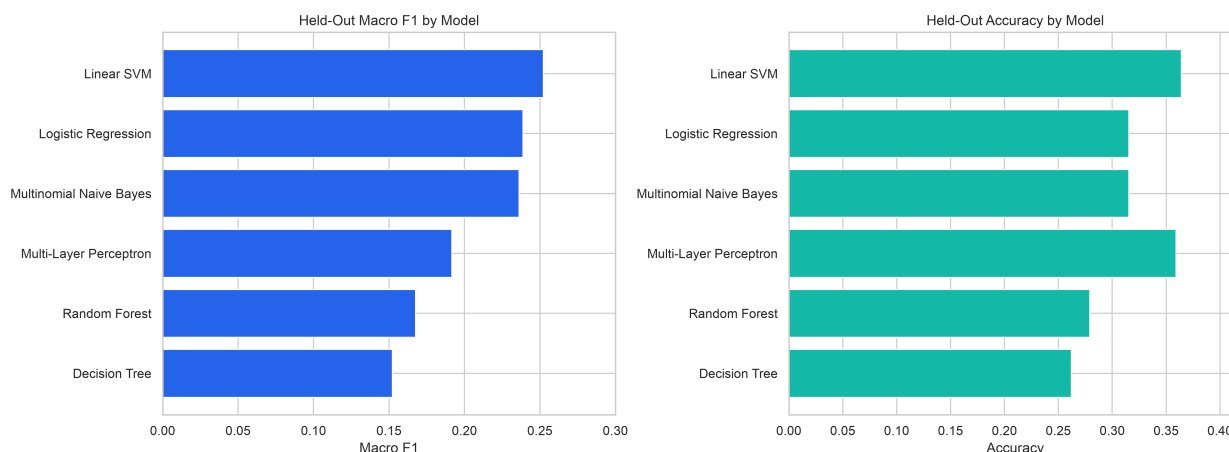
Model	Core rationale	Main tuned parameters
Logistic Regression	Sparse linear baseline	C, class weight, n-grams, selected features
Linear SVM	Maximum-margin sparse text classifier	C, class weight, n-grams, selected features
Multinomial NB	Non-negative probabilistic text baseline	alpha, priors, feature count, n-grams
Decision Tree	Nonlinear rule model	criterion, depth, split, class weight
Random Forest	Bagged tree ensemble	tree count, depth, sampled features, class weight
MLP	Dense nonlinear neural model	architecture, activation, alpha, learning rate

Every model used an 80:20 stratified split, random state 42, five-fold stratified cross-validation, GridSearchCV, and macro F1 as the tuning objective. The held-out test set was evaluated after tuning.

5. Evaluation and Comparison

Rank	Model	Accuracy	Macro F1	Weighted F1	CV Macro F1	Gap
1	Linear SVM	0.3641	0.2523	0.3661	0.2623	0.4450
2	Logistic Regression	0.3155	0.2387	0.3322	0.2662	0.4456
3	Multinomial Naive Bayes	0.3155	0.2362	0.3275	0.2651	0.5398
4	Multi-Layer Perceptron	0.3592	0.1916	0.3297	0.2068	0.2882
5	Random Forest	0.2791	0.1677	0.2751	0.2296	0.7148
6	Decision Tree	0.2621	0.1523	0.2565	0.1772	0.7342

Final Six-Model Comparison



Final held-out macro-F1 and accuracy comparison across six model families.

Linear SVM was selected because it produced the highest held-out macro F1 and accuracy. Logistic Regression and Naive Bayes were competitive, while the tree-based models strongly overfit dense SVD components. MLP achieved high accuracy relative to several models but weaker macro F1, showing bias toward larger classes. Disgust remained the weakest class across models.

Recommendation Demonstration

The saved Linear SVM pipeline predicts an emotion from a supplied description. Catalogue items are filtered by predicted emotion, optional genre, and minimum rating, then ranked using 75% normalized rating and 25% log-popularity. The executed demonstration predicted sadness and produced ranked drama recommendations. This is a content-based academic prototype rather than a production personalization service.

6. Ethical Considerations and Bias Mitigation

Severe class imbalance, subjective emotion annotation, possible cultural bias, and uneven genre representation can create unequal performance. False predictions may frustrate users and should never be treated as psychological assessments. Mitigation included stratification, macro-F1 reporting, balanced weights where supported, class-level metrics, transparent limitations, and no use of personal data. Future deployment should calibrate confidence, allow abstention and feedback, and monitor subgroup errors.

7. AI Tool Usage Declaration and Transparency

Tool	Support provided	Extent	Group verification
ChatGPT / Codex	Problem framing, code structuring, debugging suggestions, visualization ideas, tuning suggestions, documentation and language refinement	Moderate	All scripts and notebooks were executed. Metrics were read from generated files, models were reloaded, outputs were inspected, and students retain responsibility for understanding their work.

8. Reflections and Lessons Learned

The group learned that the modelling unit is as important as the algorithm: conflicting repeated reviews required description-level aggregation. Leakage-safe pipelines were essential because TF-IDF, imputers, scalers, feature selectors, and SVD learn from data. Macro F1 exposed weaknesses hidden by accuracy. Simple sparse linear models generalized better than more complex dense models on the limited dataset, demonstrating that complexity does not guarantee performance.

9. References

1. Fahad Rehman. IMDb Movie Reviews - Genres, Descriptions and Emotions. Kaggle.
2. Pedregosa et al. Scikit-learn: Machine Learning in Python. JMLR, 2011.
3. Salton and Buckley. Term-weighting approaches in automatic text retrieval, 1988.
4. SLIIT IT2011 Assignment Specification, Final Evaluation - Documentation, and Final Evaluation - Implementation, 2026.
5. Project dataset audit, data dictionary, member responsibilities, executed notebooks, and generated result files.